6.8 A Group of Singular Matrices

WORKSHEET

The question

STEP III 1996, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself In part (i), the identity matrix $\mathbf{I}$ is not of the form $\begin{pmatrix} a & a \\ a & a \end{pmatrix}$. So what could the identity element of S be?

STAGE 1 Closure

Covers: Part (i)

Ask yourself Write $\mathbf{M}_a$ for the matrix with all four entries a. What is $\mathbf{M}_a\mathbf{M}_b$?

STAGE 2 The group axioms

Covers: Part (i)

Ask yourself The identity matrix $\mathbf{I}$ is not in S. So what could the identity of S be? Solve $\mathbf{M}_e\mathbf{M}_a = \mathbf{M}_a$ for e.

Pause and try For inverses, solve $\mathbf{M}_a\mathbf{M}_b = \mathbf{M}_e$ for b.

STAGE 3 Singularity spreads

Covers: Part (ii)

Pause and try Suppose $\mathbf{A} \in G$ is singular, and $\mathbf{B}$ is any element of G. Write $\mathbf{B} = \mathbf{AC}$ for some $\mathbf{C}$ in G, using the inverse of $\mathbf{A}$ *in the group*. Take determinants.

STAGE 4 A 3×3 example

Covers: Part (ii)

Ask yourself Can you generalise part (i) to 3×3 matrices? Or find something simpler, using a matrix that ignores two of the three coordinates?

What's really going on

Idempotents and hidden invertibility

The identity $\mathbf{E}$ of any matrix group satisfies $\mathbf{E}^2 = \mathbf{E}$: it is *idempotent*. Geometrically, an idempotent matrix is a *projection*: it maps every vector onto a subspace and leaves the vectors in that subspace alone. In part (i), $\mathbf{E}$ projects the plane onto the line $y = x$, and every matrix $\mathbf{M}_a$ first projects onto that line and then stretches along it by the factor $2a$ (Figure 6.8.1). On the line, the group acts by invertible stretches; it simply ignores the direction perpendicular to it. So S is really the group of non-zero real numbers under multiplication (via $\mathbf{M}_a \mapsto 2a$), living on a line inside the plane. This is true in general: every group of matrices is, after a change of basis, a group of invertible matrices acting on the image of its identity element, padded with zeros.

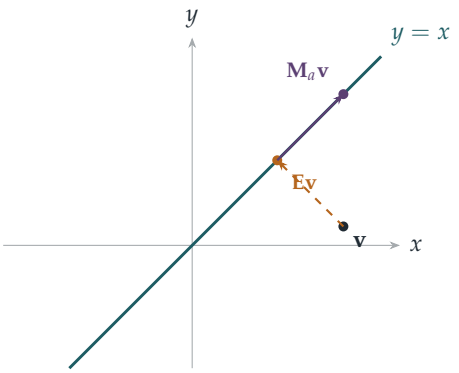


Figure 6.8.1 What the matrices in part (i) do. The identity element $\mathbf{E}$ projects every vector $\mathbf{v}$ onto the line $y = x$ (amber, dashed). Each $\mathbf{M}_a$ does the same projection and then stretches along the line by the factor $2a$ (plum; here $2a = \frac{16}{9}$). On the line itself, the group acts by ordinary invertible scaling.

Abstract and concrete

The first groups people studied were concrete: permutations, symmetries, invertible matrices. The abstract definition (a set with an operation satisfying three axioms) was distilled from them. This question shows the value of the abstract view: S and the group of non-zero reals under multiplication are *isomorphic*, the same group in different clothing, even though one consists of singular matrices and the other of numbers. Recognising when two different-looking structures are "the same" is one of the central skills of algebra, and the notion of isomorphism is how mathematicians make that precise.

Try this next

1. **An isomorphism.** Show that the map $\mathbf{M}_a \mapsto 2a$ is a one-to-one correspondence from S to the non-zero reals that turns matrix multiplication into multiplication of numbers.

2. **Not a group.** Show that the set of matrices $\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}$ with $a \neq 0$ is closed under multiplication but is not a group. (*Hint:* look for an identity element. Is there a left identity? A right identity?)

3. **Idempotents.** Find all 2×2 real matrices $\mathbf{E}$ with $\mathbf{E}^2 = \mathbf{E}$. Describe them geometrically.

4. **All entries equal.** Show that the $n \times n$ matrices with all entries equal to $a \neq 0$ form a group under multiplication, and find its identity element.

Further reading

- **M. A. Armstrong, *Groups and Symmetry*** (Springer). Matrix groups and isomorphisms, with many geometric examples.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Projections and idempotent matrices, clearly explained.
- **Joseph Gallian, *Contemporary Abstract Algebra*** (CRC Press). A friendly first course with many examples of unusual groups.

The full guided solution is in Section 6.8:
https://beyondstep.keithmcnulty.org/ch06/singular-group.html